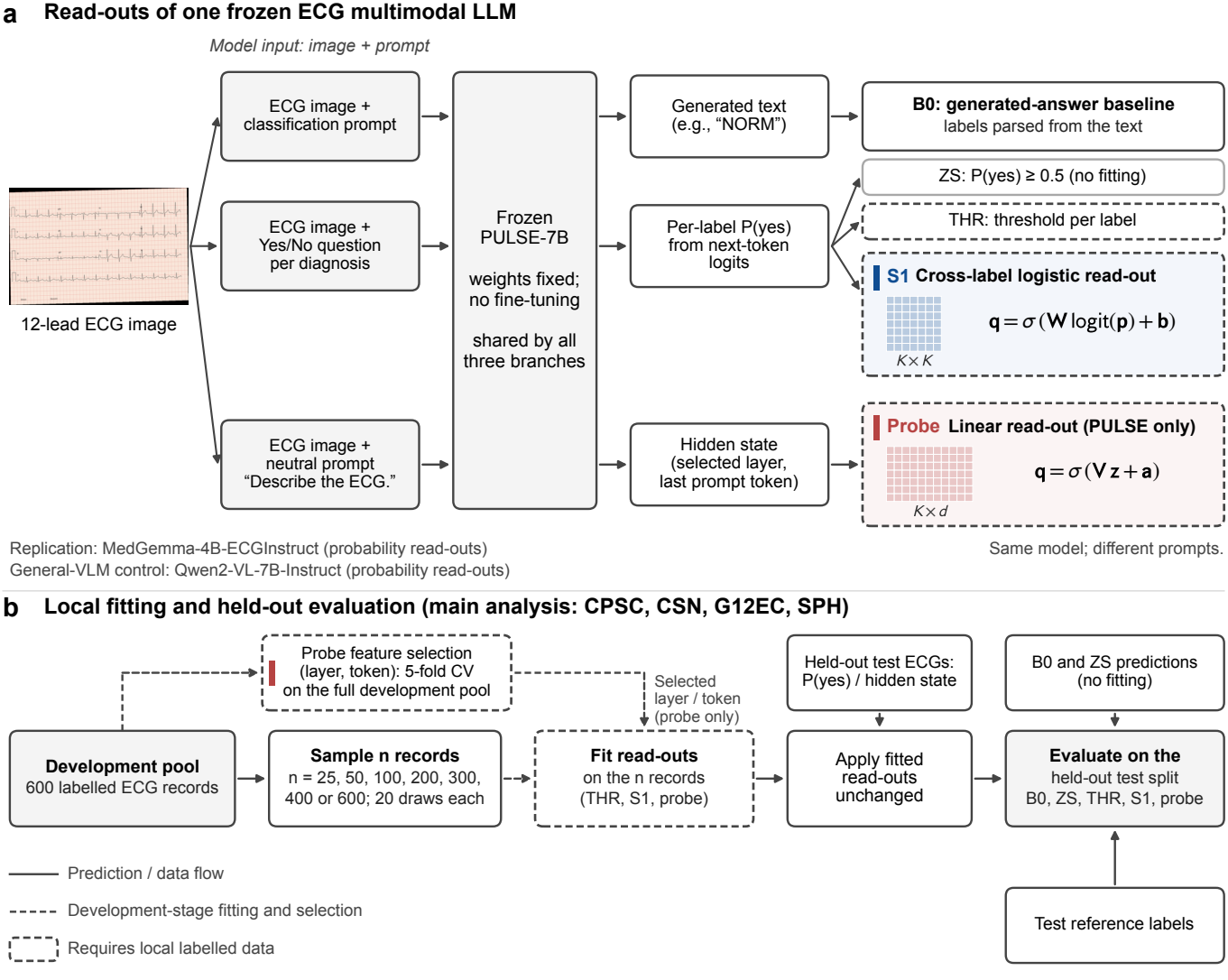

a
Read-outs of one frozen ECG multimodal LLM
Model input: image + prompt
12-lead ECG image
ECG image + classification prompt
ECG image + Yes/No question per diagnosis
ECG image + neutral prompt "Describe the ECG."
Frozen PULSE-7B
weights fixed; no fine-tuning
shared by all three branches
Generated text (e.g., "NORM")
Per-label P(yes) from next-token logits
Hidden state (selected layer, last prompt token)
B0: generated-answer baseline
labels parsed from the text
ZS: P(yes) ≥ 0.5 (no fitting)
THR: threshold per label
S1 Cross-label logistic read-out
$\mathbf{q} = \sigma(\mathbf{W}\,\mathrm{logit}(\mathbf{p}) + \mathbf{b})$
K×K
Probe Linear read-out (PULSE only)
$\mathbf{q} = \sigma(\mathbf{V}\,\mathbf{z} + \mathbf{a})$
K×d
Replication: MedGemma-4B-ECGInstruct (probability read-outs)
General-VLM control: Qwen2-VL-7B-Instruct (probability read-outs)
Same model; different prompts.
b
Local fitting and held-out evaluation (main analysis: CPSC, CSN, G12EC, SPH)
Development pool
600 labelled ECG records
Probe feature selection (layer, token): 5-fold CV on the full development pool
Sample n records
n = 25, 50, 100, 200, 300, 400 or 600; 20 draws each
Selected layer / token (probe only)
Fit read-outs
on the n records (THR, S1, probe)
Held-out test ECGs: P(yes) / hidden state
Apply fitted read-outs unchanged
B0 and ZS predictions (no fitting)
Evaluate on the held-out test split
B0, ZS, THR, S1, probe
Test reference labels
Prediction / data flow
Development-stage fitting and selection
Requires local labelled data